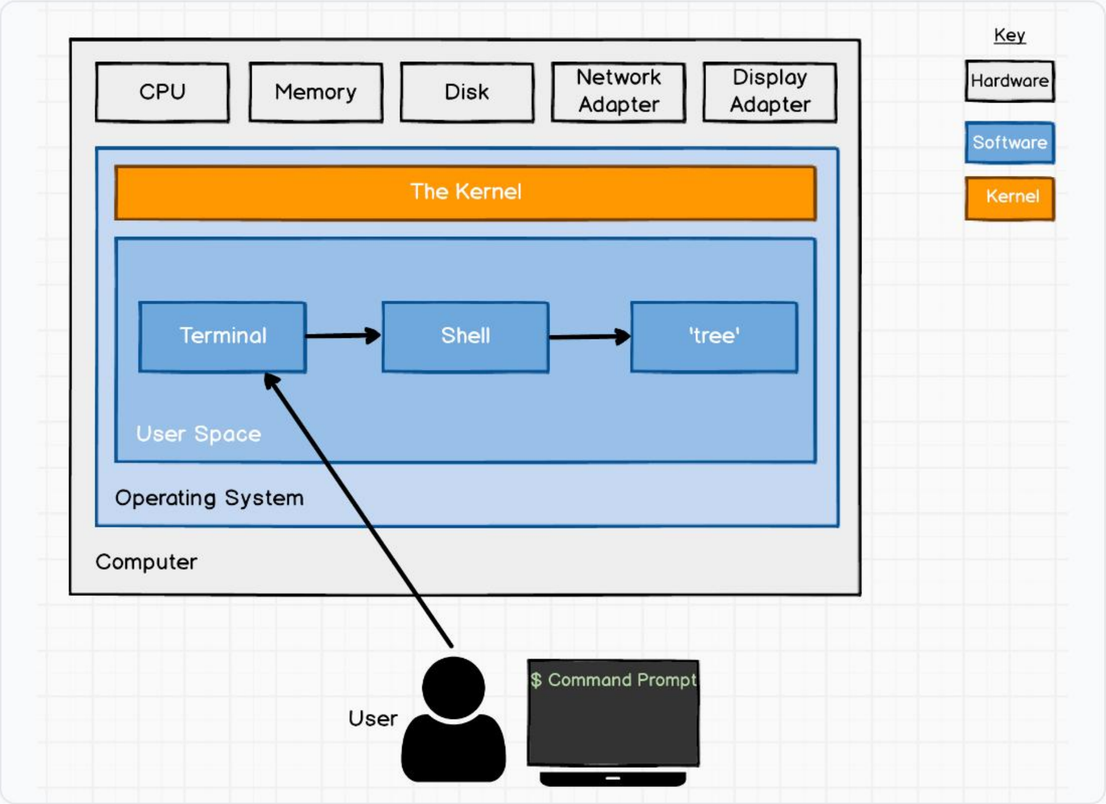
# 21. The Shell

### The Kernel's Protector

The outermost layer of an OS providing a user interface to kernel services. It "wraps" the kernel like a protective layer.

**The Function:** Translates user commands (typing or clicking) into instructions for the kernel.

*Example:* Graphical shells use icons/folders, while command-line shells (Bash, Zsh) use text commands like `cd Documents`.



# 22. CLI

### Direct Power & Control

A text-based interface used to interact with software by typing discrete lines of text.

It provides a powerful way to control a computer without graphical overhead, allowing for automation and scripting.

**Example:** Instead of clicking "Delete", a pro simply types `rm file.txt`.

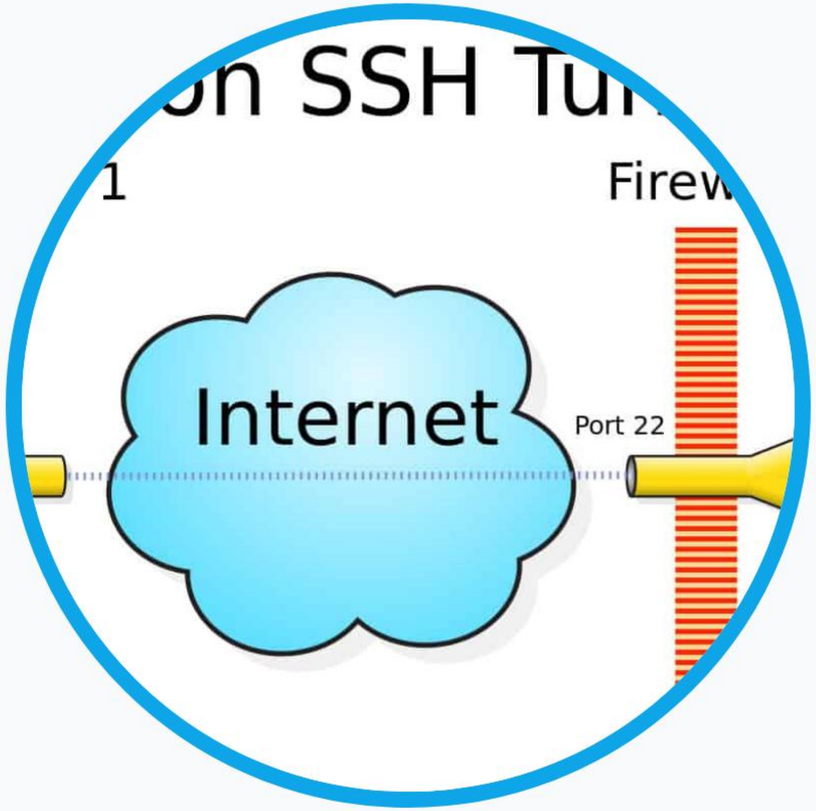
```
 ping -q fa.wikipedia.org
imedia.org (208.80.152.2) 56(84) bytes of data.

media.org ping statistics ---
ed, 1 received, 0% packet loss, time 0ms
v = 540.528/540.528/540.528/0.000 ms
 pwd

 cd /var
]# ls -la

 root 4096 Jul 30 22:43 .
 root 4096 Sep 14 20:42 ..
 root 4096 May 14 00:15 account
 root 4096 Jul 31 22:26 cache
 root 4096 May 18 16:03 db
 root 4096 May 18 16:03 empty
 root 4096 May 18 16:03 games
 gdm  4096 Jun  2 18:39 gdm
 root 4096 May 18 16:03 lib
 root 4096 May 18 16:03 local
 root   11 May 14 00:12 lock -> ../run/lock
 root 4096 Sep 14 20:42 log
 root   10 Jul 30 22:43 mail -> spool/mail
 root 4096 May 18 16:03 nis
 root 4096 May 18 16:03 opt
 root 4096 May 18 16:03 preserve
 root 4096 Jul  1 22:11 report
 root    6 May 14 00:12 run -> ../run
 root 4096 May 18 16:03 spool
 root 4096 Sep 12 23:50 tmp
 root 4096 May 18 16:03 yp
]# yum search wiki
gpacks, presto, refresh-packagekit, remove-with-leaves
tes                                                          | 2.7
tes/primary_db                                               | 206
pdates                                                       | 2.7
                                                             | 5.9
                                                             | 4.7
                    73% [=======================        ]  62 kB/s | 2.6
```

# 23. SSH (Secure Shell)




### Encrypted Connectivity

A cryptographic network protocol for operating services securely over unsecured networks.

It creates an encrypted "tunnel" between a local machine and a remote server, protecting passwords and commands.

**Usage:** A dev in NY updating a server in London safely via "terminal teleportation."

# 24. Mainframe

**99.999%**

**Five-Nines Availability**

### Massive Throughput

High-performance computers designed to process billions of transactions in real-time with extreme reliability.

Unlike supercomputers (math-focused), mainframes focus on millions of simultaneous small tasks (like bank transfers).

*Example:* Swiping a credit card triggers a mainframe check in milliseconds.

# 25. Programming Language

### Human-Hardware Bridge

A formal set of rules and syntax used to write instructions for a computer.

It allows developers to express complex logic using words (if, while, print) instead of raw binary (0s and 1s).

**Modern Logic:** You write `console.log("Hello")` and the language handles the translation to hardware.

# 26. Abstraction

### Complexity Management

Hiding complex, messy details behind a simple interface. Focus on *what* it does rather than *how*.

### The Car Analogy

To drive, you only need the wheel and pedals. You don't need to understand fuel injection to travel.

---

In computing, programming languages are abstractions of the underlying silicon and electricity.

# 27. Interpreted Language

### Execution

Source code is executed directly, line-by-line, by an external program at runtime.

### Agility

Faster testing and development; no long "build" process required before running.

### Examples

Python and JavaScript. If line 10 has a bug, the program crashes exactly when it hits that line.

# 28. Interpreter

### On-the-Fly Translation

A program that executes instructions directly without prior compilation into machine code.

It reads source code, parses it, and performs actions simultaneously. No standalone binary is created.

**The UN Analogy:** Like a live translator at a conference—speaking the translation as the original speaker talks.

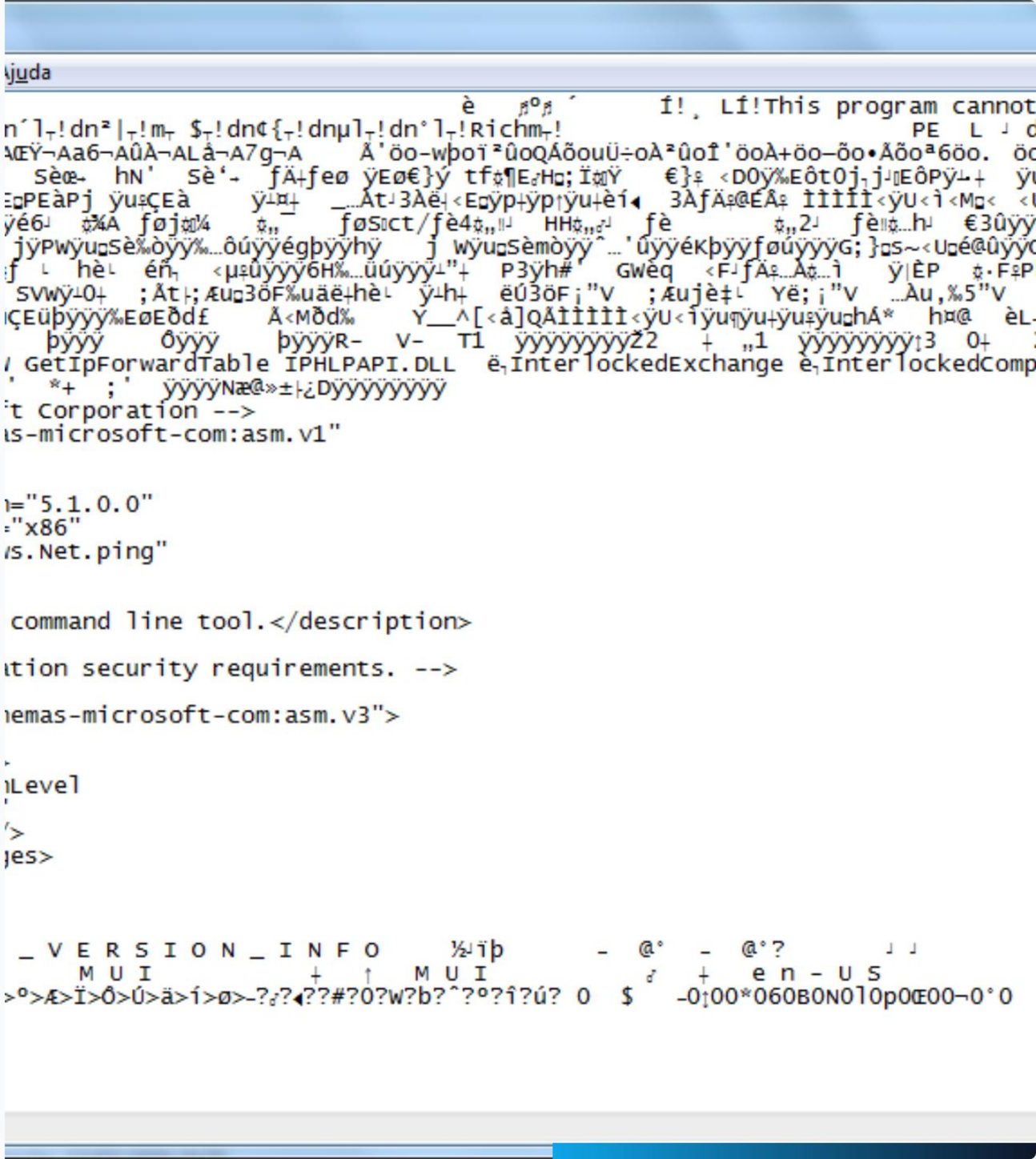
# 29. Compiled Language

### Pre-Translated Speed

Source code is translated into binary by a specialized program *before* execution.

Produces a standalone "executable" file. Because translation is finished, these languages are incredibly fast.

**Rigidity:** Typo anywhere? The build fails. Examples: C++, Rust.



# 30. Compiler

### Process

Translates high-level source code into machine binary all at once before usage.

### Analogy

Like a book translator—they translate the whole manuscript into a final book. Reader doesn't need the original.

Compilers optimize code for specific CPU architectures and check for syntax errors across the whole project.

# 31. Executable

### Windows

Typically ends in **.exe**

### macOS

Found in **.app** bundles

### Linux

Files with **execute** permission

Pre-translated binary instructions ready for the CPU to run immediately upon opening.

# 32. Variable

### Labeled Containers

A named storage location in memory holding a value. It allows programmers to store and retrieve data easily.

Instead of memory addresses (0x7f...), we use names like `userAge`.

**Example:** `score = 10`. If the player wins, update to `score = 20`. The name stays, the content changes.

# 33. Data Type

A classification telling the computer how to interpret and use a value, determining memory allocation and legal operations.

| Type | Classification | Example |
|---|---|---|
| **Integer (int)** | Whole Numbers | 5, -42 |
| **String (str)** | Text | "Hello" |
| **Boolean (bool)** | Logic | True, False |
| **Float** | Decimals | 3.14 |

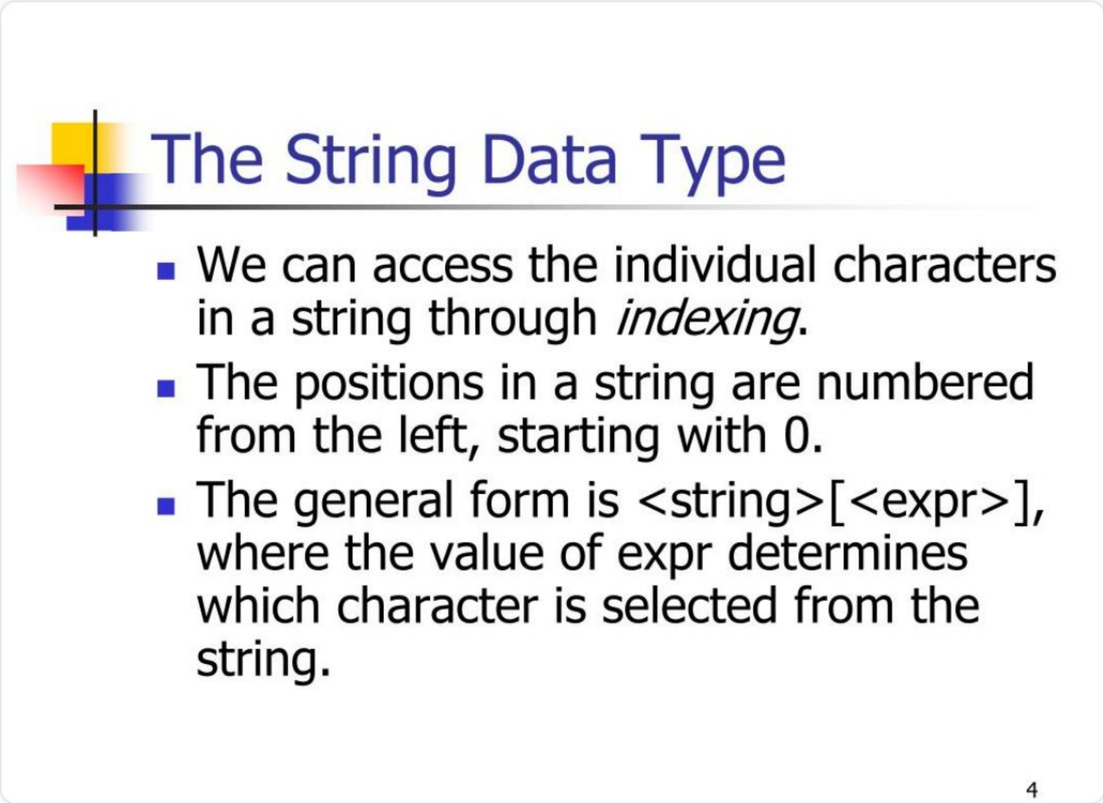
# 34. String

### Literal Text Sequences

A data type representing text as a sequence of characters bundled together.

Wrapped in quotes (" ") to distinguish it from commands or variables. Even numbers in quotes ("123") are treated as text.

**Purpose:** Allows computers to process human language and communicate with users.



# 35. Pointer

### The Memory Map

A special variable that stores the memory address of another variable rather than a direct value.

Allows direct manipulation of memory, crucial for complex data structures and resource management.

**Analogy:** Like a treasure map—it isn't the gold, but the coordinates showing where the gold is buried.

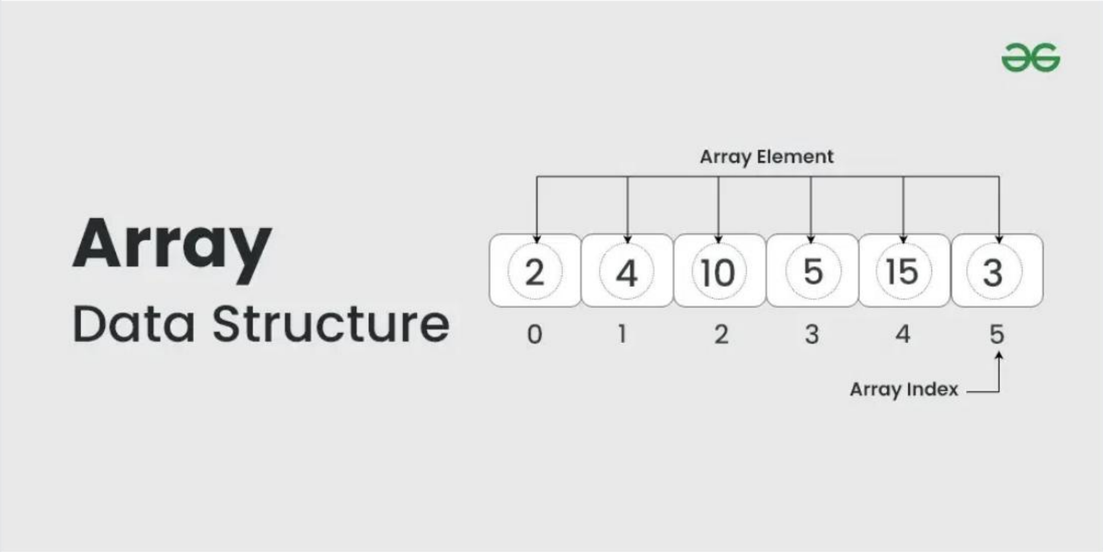
# 36. Array

### Contiguous Block Data

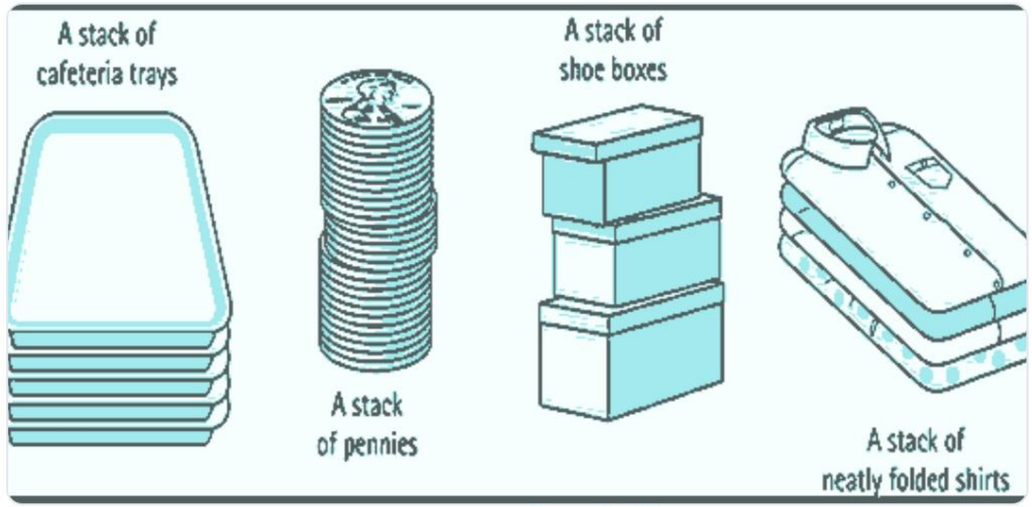
A data structure storing a collection of elements (usually same type) in a single block of memory.

Items are accessed via an **index**. Computers almost always start counting at zero!

**Efficiency:** Grouping `highScores[0..n]` is far better than individual variables.



# 37. Stack (LIFO)




## LIFO Principle

Last-In, First-Out. Like a stack of cafeteria trays.

## Operations

**Push:** Add to top
**Pop:** Remove from top

Used for "Undo" buttons and function calls.

# 38. Queue (FIFO)

## FIFO Principle

First-In, First-Out. Like a line at a coffee shop.

## Operations

**Enqueue:** Add to back
**Dequeue:** Remove from front

Used for printer tasks and data packets.

# 39. Object (OOP)

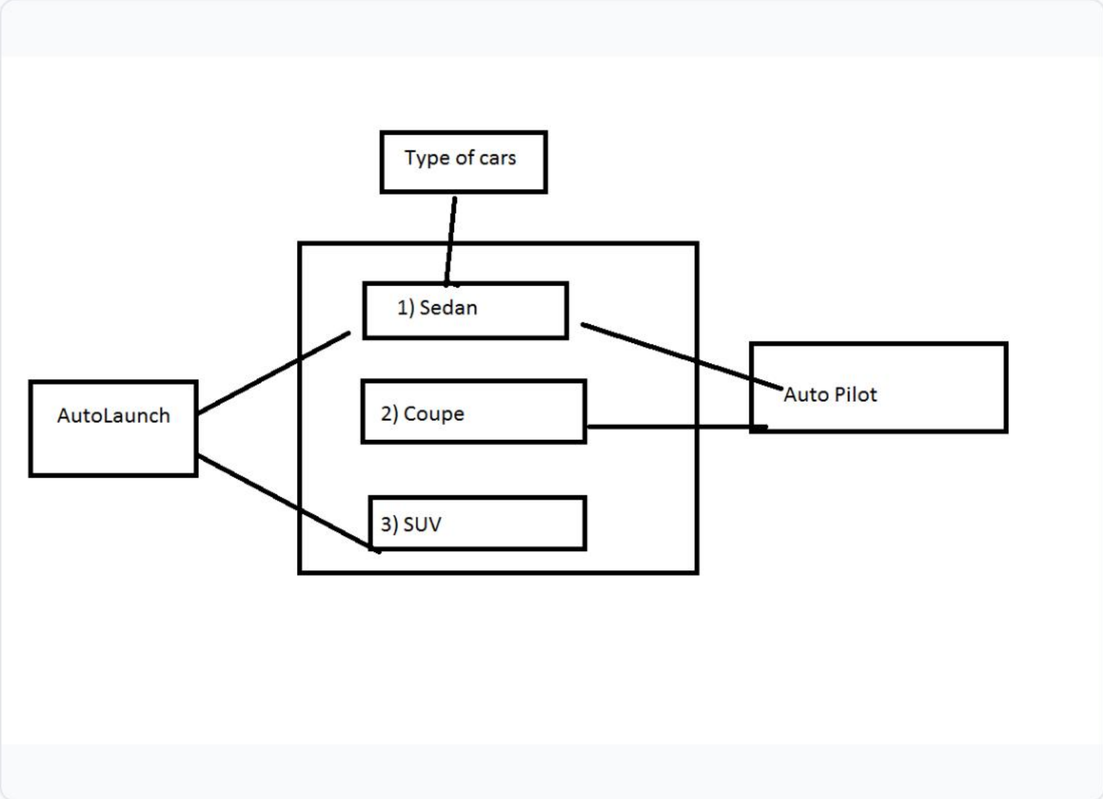
### Self-Contained Units

In OOP, an object bundles data (properties) and logic (methods) together.

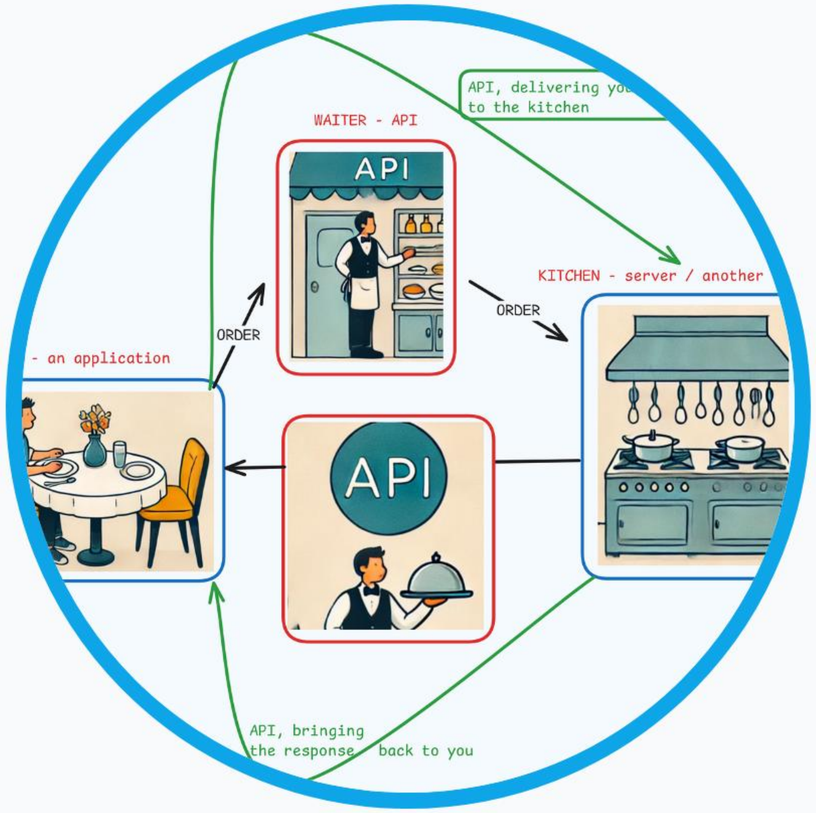
It models real-world entities. You group everything related to one thing into one unit.

**Car Object Example:**

- **Properties:** color: "Red", speed: 60
- **Methods:** `brake()`, `accelerate()`



# 40. API




## Digital Contract

Rules and protocols allowing different apps to communicate. A "bridge" between systems.

Allows using third-party services (Maps, Payments) without knowing their internal code.

**The Waiter Analogy:** You order via the menu (API Doc); the waiter (API) gets food from the kitchen (Server).

# Image Sources

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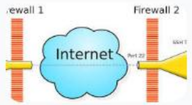


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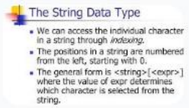


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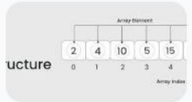


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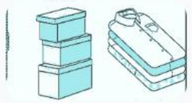

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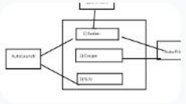

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